

Market Moods: Linking S&P 500 Swings With Sentiment Scores

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Figure 1: S&P500 Graph for Fiscal Year 2024, Yahoo Finance

Abstract

This project explores the relationship between sentiment in news and social media and stock market trends, specifically the S&P 500. For our milestone, we collected and analyzed financial sentiment from Twitter, Reddit, and major news sources (both liberal and conservative) using web scraping tools like Selenium and BeautifulSoup, among others. For our post-milestone work, we used the NLTK model to gain a sentiment analysis for the data we collected, and analyzed the correlations with the market. The sentiment trends

were correlated with S&P 500 movements to assess predictive capabilities, where negative scores are associated with predicting downwards movements, and vice versa for positive scores predicting upwards movements. These results can be seen in the graphs attached at the end of this report. By combining sentiment analysis with market data, we aim to identify patterns that could improve stock market forecasting, and assess how useful a particular data source to decide how the stock market was going to perform relatively from one month to the next.

0.1 Github Repository

This is important, below you will find the link to the GitHub repository consisting of our code for determining our analysis.

github.com/uparkh/news-stock-predictions

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1 Introduction

For the final submission, our focus was on processing data and analyzing results from Youtube, Reddit, and both liberal and conservative news outlets to analyze sentiment for stock market prediction. We aimed to efficiently process and store relevant sentiment data. We classified sentiment using VADER as 1: positive, 0: neutral, -1: negative. While social media data collection proceeded as expected, we encountered challenges when extracting news articles due to site restrictions and inconsistencies in formatting. These obstacles required us to adjust our approach, refining our selection of sources and modifying our data extraction methods to ensure reliable sentiment analysis.

As a result, we revised our expectations regarding data availability and adjusted our collection strategy to maintain the integrity of our analysis. Despite these hurdles, we successfully gathered a substantial portion of the required data and refined our preprocessing pipeline to standardize text input for sentiment classification. Moving forward, we will focus on optimizing our sentiment scoring model and establishing correlations with S&P 500 movements.

2 Experimentation

2.1 Data Preprocessing

The majority of the preprocessing involved: tokenizing, removing numbers, removing special characters, and lowercasing the text. But one implementation decision we had to make was whether to use a stemmer or lemmatizer approach to convert words into their root form. We ran an experiment and tried both out on a sample sentence from one of our Reddit datasets.

We chose this sample sentence describing a story.

Parents scammed out of 30k in crypto As the title says my parents got scammed out of 30k in crypto but we're not sure how exactly the guy got access to their wallets. My dad downloaded the app Zoho on his phone but his Coinbase wallet was on his computer and somehow the guy gained access to all of his wallets on his computer without ever downloading zoho onto his actual computer. How is this possible? My dad never gave them his usernames or passwords or seeds or anything and somehow the guy got into their account. Has anybody had experience with this?

Our stemmer (NLTK's PorterStemmer) preprocessing output was:

parent scam k crypto titl say parent got scam k crypto sure exactli guy got access wallet . dad download app zoho phone coinbas wallet comput somehow guy gain access wallet comput without ever download zoho onto actual comput . possibl ? dad never gave usernam password seed anyth somehow guy got account . anybodi experi ?.

Whereas our lemmatizer (NLTK's WordNetLemmatizer) preprocessing output was:

parents scammed k crypto title say parent get scammed k crypto sure exactly guy get access wallet . dad download app zoho phone coinbase wallet computer somehow guy gain access wallet computer without ever

download zoho onto actual computer . possible ? dad never give usernames password seed anything somehow guy get account . anybody experience ?.

As you can see, the stemmer can frequently produce nonsense words because it blindly applies heuristics / rules for obtaining word stem without paying attention to different parts of speech of the sentence. By combining Part-Of-Speech Tagging with a lemmatizer, the lemmatizer's output produced words that were much more sensible. We figured that this would be important, seeing as we are not going to be training our own sentiment analysis model, and instead using a pretrained model. We wanted to feed in words that are actually in the dictionary, so the pretrained models do not get tripped up. Thus we chose to lemmatize.

Note that keeping the punctuation was intentional. After much experimentation we discovered that it did not really trip up the VADER output, but was necessary for FinBERT to even have any sensible output. See next section.

2.2 Sentiment Model Selection

The primary issue now was (1) how we were going to organize the data to be fed into the models, and (2) which model to select. We wanted to experiment with how different NLTK's VADER or FinBERT output would be. VADER is more optimized for general texts, including social media posts, while FinBERT is better optimized for formal financial texts. We experimented primarily on the Reddit datasets.

The first thing we tried was to concatenate *all* of the text columns for a given month, into one giant "month corpus." We ended up with 12 "month corpora" where each corpus contained tens of thousands of characters. Needless to say, this did *not* work very well. VADER classified each corpus as extremely positive, and no useful output even when standardized. We also discovered at this stage that FinBERT can only take 512 tokens at a time.

So our second approach was to take the $M = 11$ preprocessed columns, and for *each individual* post i : perform sentiment analysis with both VADER and FinBERT on the "headline" s_0 (post title concatenated with selftext), and for each of the top 10 comments $s_1, s_2, \dots, s_{10}$ on the post. Then take a weighted mean:

$$\mu_i = \sum_{j=0}^{M=11} w(j) * s_j$$

Where $w(j) = e^{-3i/M} = e^{-3i/11}$. This weight function isn't exactly a probability distribution, but the idea is to really emphasize the sentiment of the first few text columns). The formula was chosen by hand, empirically, to make sure the first few columns get prioritized, but to have some decent weight to the last few columns.

Finally, after getting a proper sentiment score for all N posts, we now have a vector $\vec{\mu} = \langle \mu_0, \mu_1, \dots, \mu_N \rangle$ of each post's weighted mean sentiment. Each post is tagged by month, so we averaged all the scores for each month's posts' μ -scores. So we now have 12 months sentiment scores for 2 models. For each model, we standardized the scores across the 12 months, because we were not interested in the raw scores, but rather the relative changes in raw score. As we discussed in class, one source may be in general more pessimistic or optimistic than another, so this step was necessary. Here were our experimental results:

Month	NLTK VADER	FinBERT
Jan	1.532864	0.573915
Feb	0.9994	0.579801
Mar	-0.83769	-0.638512
Apr	-0.587399	0.451187
May	1.44083	0.1298885
Jun	0.169872	-0.981853
Jul	-0.080389	0.422455
Aug	-1.528301	-0.69398925
Sep	-0.917136	-0.09897425
Oct	-0.407941	0.1118875
Nov	0.858576	0.3077205
Dec	-0.642686	-0.163526

Table 1: Standardized monthly sentiment scores using VADER and FinBERT.

It seemed that the polarities for the FinBERT relative monthly changes were more muted than VADER, which graded sentiment more aggressively with higher magnitude swings. Perhaps FinBERT was indeed more "accurate," but for our analysis, VADER was significantly faster to run, and provided reasonable output upon manual inspection of data. FinBERT took around 20-40 minutes for some subreddits, while VADER only 1-2 minutes. Since we changed around our datasets based on various challenges quite a bit, we went with VADER.

This calculation procedure for the Reddit dataset with $M = 11$ preprocessed columns can now take in any dataset with any number of M preprocessed columns, so now that we experimented and found the best procedure for our use case, we applied it to all the datasets we collected.

2.3 Graph Creation and Output

So now that we tackled 2 main implementation challenges, with choosing the lemmatizer and NLTK VADER model, we lastly needed to deal with how to actually represent the S&P500's 2024 performance. As shown on the first page of the paper in Figure 1, the absolute price will always have an average trend up and to the right because the stock market always has some average annualized return.

To account, for this, we took the trendline connecting the start and end closing prices of the year, and then subtracted this trendline from the absolute price to get a deviation measure of sorts. See Figure 2 for an illustration of how the deviation adjustment was calculated.

Now we have a good way to measure relative swings in price movements, adjusting for the average increase in stock returns. We chose to then superimpose a histogram of VADER's monthly output on the deviation lines. Each of these final graphs can be found in the Appendix.

3 Analysis of Final Predictions Against Source Data

As mentioned before, all graphs that were generated using the steps discussed in the previous section are found in the [Appendix](#) of this

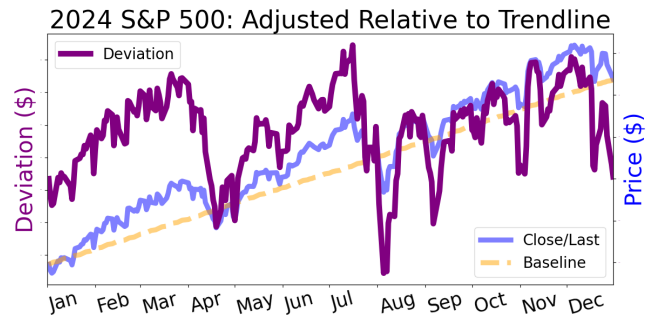


Figure 2: S&P500 Deviation

document. We will refer to our graphs by appendix sections in our individual analyses.

3.1 Various Finance Subreddits

We will not go into deep analysis here on how we collected the data, nor how it was organized, as that was covered more in detail in the milestone paper. But as discussed in sections 2 and 3 of this paper, we generated graphs that overlay the sentiment scores calculated for each of the four subreddits for each month, over the relative price movements of the S&P 500 for fiscal year 2024. These graphs can be found in Appendix A.

We discussed why we chose these subreddits in our milestone paper, but we briefly reiterate the reasons again here, to provide a refresher on what we hypothesized versus how strong the correlation was.

- **r/finance**

This subreddit is more serious, and generally consists of folks talking about general financial wellbeing, and other discourse perhaps around economics. So we hypothesized their sentiment may be more measured or mature.

In the graphs, we observed that as the the stock market performed better than the trendline, sentiment was roughly uncorrelated for the first half of the year. But the big downwards spike in August and the later upwards spike in September produced negative and positive reactions from this subreddit.

- **r/investing**

Like the previous one, but more focused specifically on investing. Obviously we want to see this subreddit's sentiment scores as this community mainly consists of those wanting to get into long-term investing.

As for the graph, we noticed that in May 2024, [AI Chip Stocks like AMD, Qualcomm, and NVIDIA surged](#), and perhaps savvy tech investors in this subreddit consequently felt optimistic about the future. Other than the reactionary negative spike in August, it seems that investors were largely mute for the rest of the year.

- **r/cryptocurrency**

This subreddit was not in our original proposal, but we added this one because while crypto is not a traditional asset, we were curious to see if sentiment scores here would correlate with the S&P500 by any proximity.

It seems that posts in this subreddit were overall disconnected from the S&P500's price deviations. Even the August downwards spike in the price deviation did not provoke a severe negative reaction. If we look at Bitcoin's performance in January 2024, and in October 2024, we noticed that there were 2 huge increases in BTC value, and suspect that is likely the reason for the positive sentiment in early 2024 and in November 2024 for this subreddit.

- **r/wallstreetbets**

This reddit is a bit more on the "meme" side of the Reddit culture and consists of a lot of retail investors who are extremely risky with their bets, almost approaching gambling. We thought this could be interesting to analyze their sentiments.

Indeed, this is funnily enough the most correlated graph of the entire bunch to the stock market movements! We believe this is because a lot of users on this subreddit like to "bet" on SPY ETF, which indexes the S&P500 directly for trading. Combine this fact with the other observation that users in this community are very risky with their money, and the graph shows extreme reactionary swings based on the price deviations of the S&P500!

For each of these analyses, we had a great time analyzing why each graph turned out the way it did. But to validate our research, we'd likely need to take a deeper dive and perhaps perform a topic classification, as discussed in class, to be able to better understand what folks were discussing during a particular month, to get a better idea of the causation of real world events to the sentiment output. This is a good direction to take further improvements on our work here.

3.2 Liberal News Outlets

For our analysis of liberal news outlets, we aimed to collect and evaluate the sentiment of articles from left-leaning media sources throughout 2024. Initially, we attempted to scrape articles from CNN and Vox, but encountered challenges due to anti-scraping mechanisms implemented on their websites, making large-scale data collection difficult.

To address this, we shifted our focus to news outlets that provided developer-friendly access to their content. We discovered that The New York Times offers an Archive API, which allowed us to retrieve metadata and article text dating back to the 1800s. Leveraging this API, we successfully gathered data for approximately 50,000 NYTimes articles published in 2024.

After compiling the dataset, we processed it through our sentiment analysis model, generating the results visualized in Appendix C. The graph indicates a notable correlation between the sentiment scores of NYTimes articles and stock price trends, particularly at

the beginning of each month. However, while the alignment is observable, we refrain from asserting definitive accuracy in sentiment scoring due to inherent limitations in our analysis.

3.3 Conservative News Outlets and Yahoo Finance

To process the data from the json format to the csv format our group agreed for the *raw data to graph.py* batch job, a script was written in the yahoo directory. This script: *json extract.py* was pointed to a directory where it extracted all json files and concatenated them into one file with the format of *date, title, text0, text1, ..., text_n*. Where *text_n* is the *n_{th}* sentence. This format was then fed into our model to produce the graph. A command line argument was added to the script to allow changing the name of the output png image.

From our graph generated we can see periods of strong positive sentiment from February-March, July, and November often coincide with or slightly precede upward deviations in the S&P 500, suggesting some short-term correlation between positive news sentiment and market optimism.

We also see negative sentiment periods May-June, August-October appear to align with or precede dips in the S&P 500's deviation from trendline, suggesting possible pessimism or market correction following news negativity. However, the correlation is not perfect. For example, during late October to November, sentiment turned positive again, but the price deviation remained relatively flat or declined, highlighting the noisy nature of market behavior.

These observations suggest that sentiment data from Yahoo Finance may hold some predictive value, particularly in identifying potential inflection points. However, the signal appears to be weak to moderate at best and should likely be taken with a grain of salt. The utility of Yahoo Finance sentiment as a leading indicator is promising, but requires further statistical validation.

3.4 Twitter

3.4.1 From X (Twitter) To Youtube.

To analyze financial sentiment on Twitter, we used the Twitter API (Tweepy) to scrape tweets containing different keywords related to finance. These keywords were "finance", "stock", and "investing". The script included retry logic and rate limiting, which allows us to gather large amounts of data. For the milestone, we chose to only display 100 points of data since at this moment, we are not analyzing the data. Using the API, we are also able to get metadata such as timestamp, likes, retweets, hashtags, etc. This will help us contextualize the analysis. Skimming the data, Twitter seems to have a high engagement level around cryptocurrency. Furthermore, unverified accounts dominate the data, which we can decide to filter out for more concrete data. Next steps for the Twitter involve testing with a model such as BERT to make the search keywords more accurate according to S&P500 movements. Overall, the implementation was very successful and allows for easy customization when compared with the alternative, web-scraping.

Initially, we planned to analyze Twitter due to its real-time nature and high volume of public commentary. However, after the milestone phase, we realized an issue: the Twitter API we used only allowed access to tweets from the past 7 days. Since our project

required data spanning the entirety of 2024—and ideally the past few years—this limitation made Twitter unusable for our goals.

To work around this, we tried web scraping instead. Unfortunately, Twitter has multiple anti-crawling measures that make scraping virtually impossible. These include dynamic content loading through JavaScript, rate limiting and IP throttling, and requiring login tokens and session cookies.

Wanting to stick with a popular platform, we shifted focus to YouTube. Unlike Twitter, YouTube's Data API allows access to public video comments and metadata across a much bigger timeframe. Since most of user-written text on YouTube is found in the comments, we used this as the data source. There are many irrelevant comments, so we filtered by top financial news providers (CNBC, Bloomberg, Yahoo Finance), and retrieved the top comments for each video for each month in 2024 using the YouTube Data API with keyword filters and comment ranking retrieval.

After processing the comments through our sentiment analysis pipeline, we found that the sentiment expressed in YouTube comments mostly diverges from the actual S&P500 market. Looking at a few of our retrieved comments, it can be seen that a major issue is many comments focus on the production quality or entertainment value of the video itself. Also, comments tend to reflect personal opinions or reactions to the presenter, not to the underlying financial data or trends being discussed.

Using the sentiment score against the S&P 500, we found little correlation. In several cases, the spikes in sentiment had no corresponding impact on the market. Although YouTube remains a valuable source of public opinion, we conclude that its comments are not a strong proxy for financial market sentiment, at least not in their raw form.

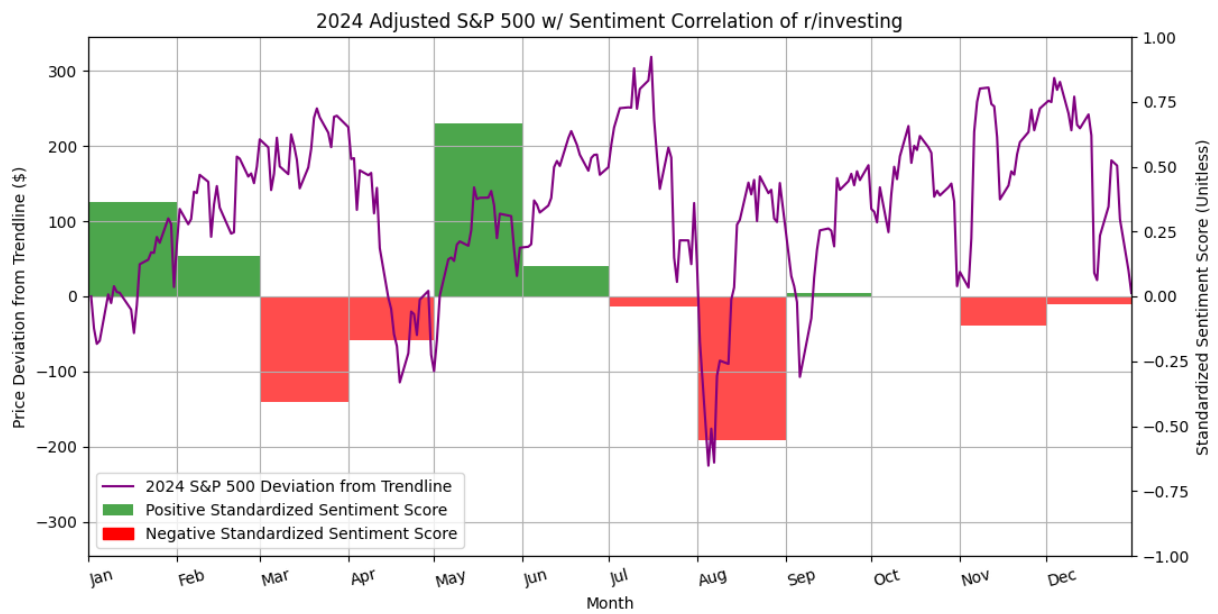
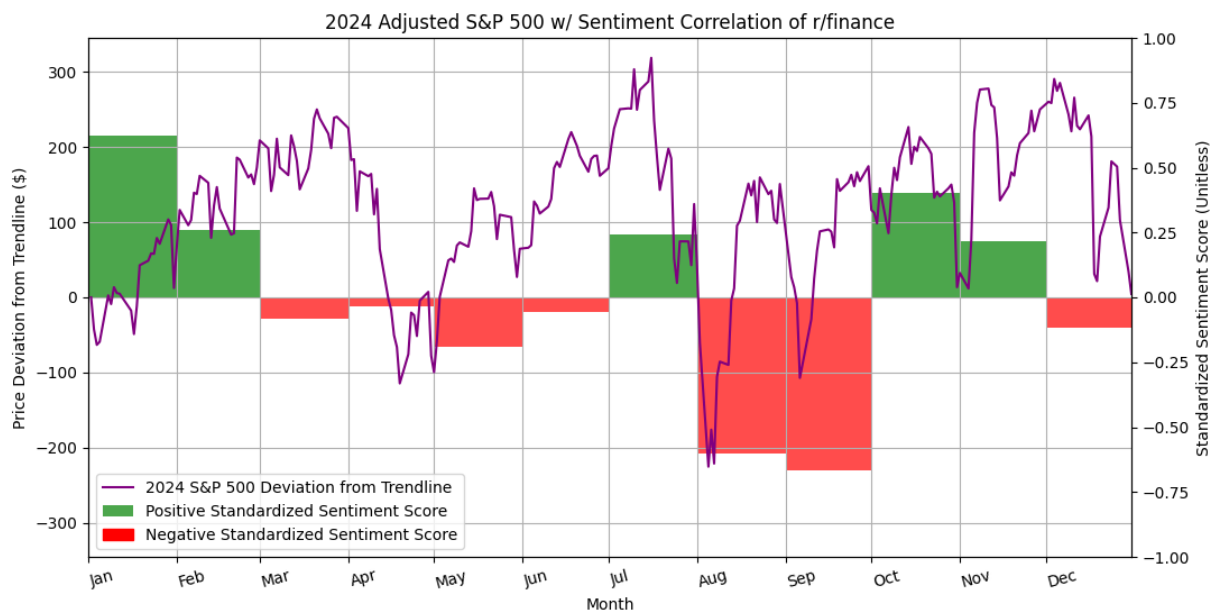
4 Discussion, Further Work

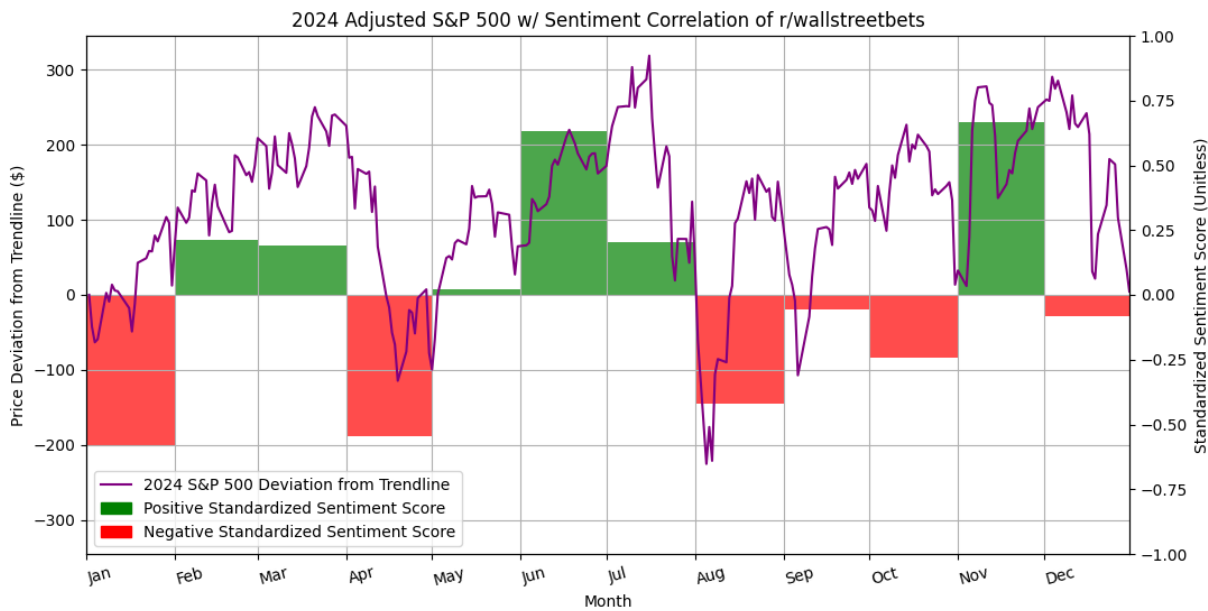
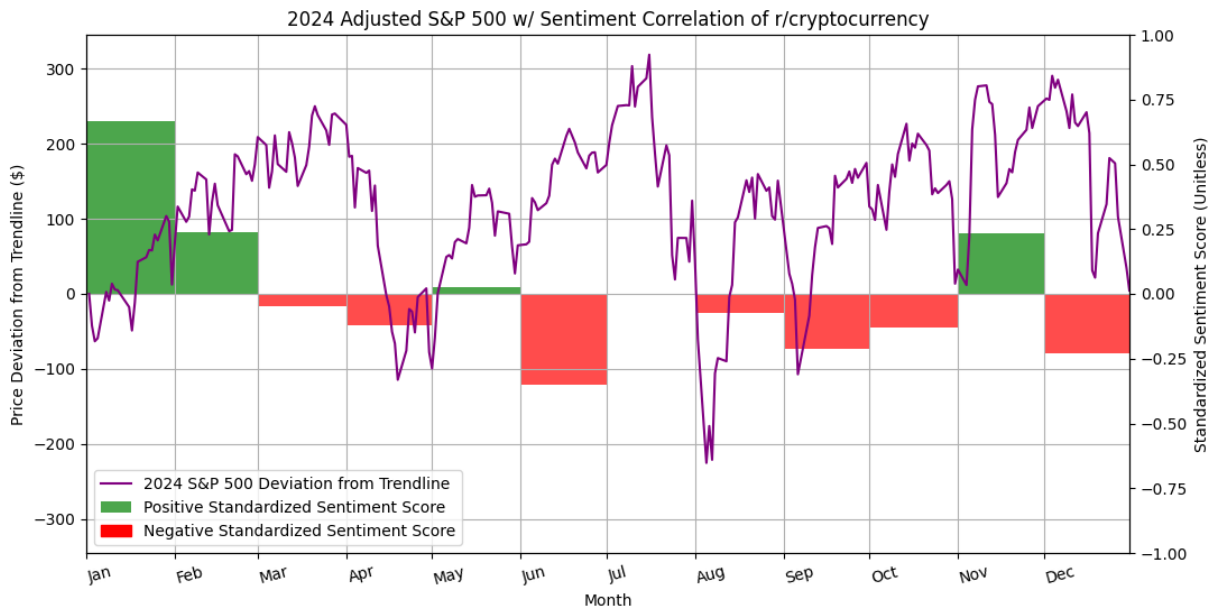
Overall, using Vader for sentiment analysis of several sources to estimate S&P500 performance proved to be somewhat successful. Further work would include a more focused analysis of different markets with an interval size smaller than 1 month. Also, many of our data may be reactionary, so testing predictability would be a logical next step.

5 Disclaimer

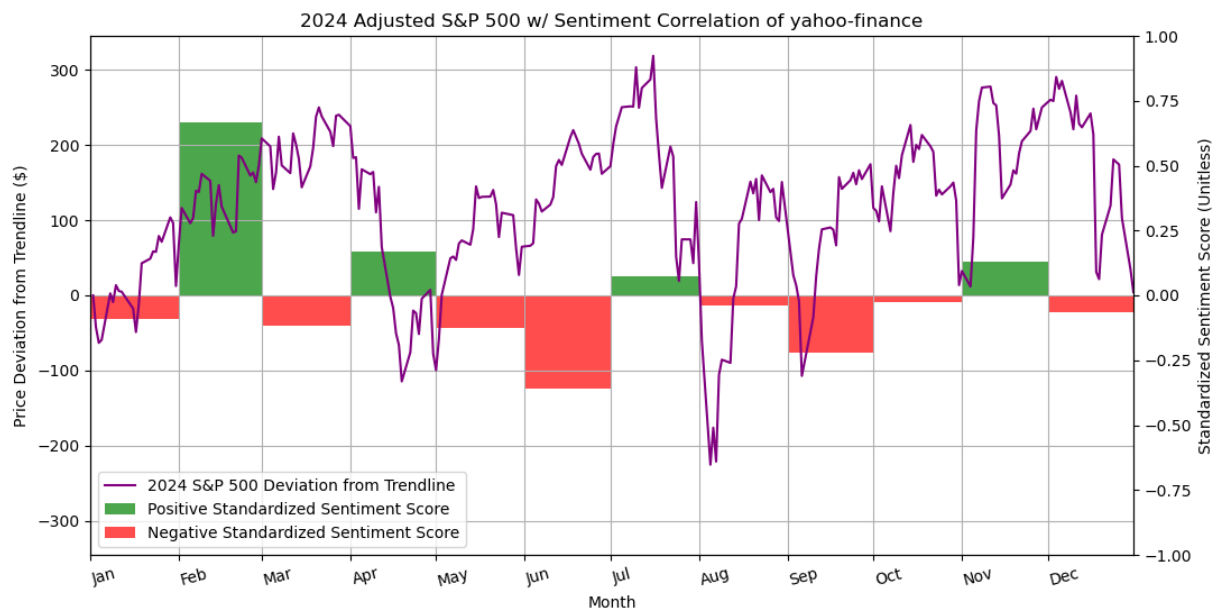
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A Reddit Graphs

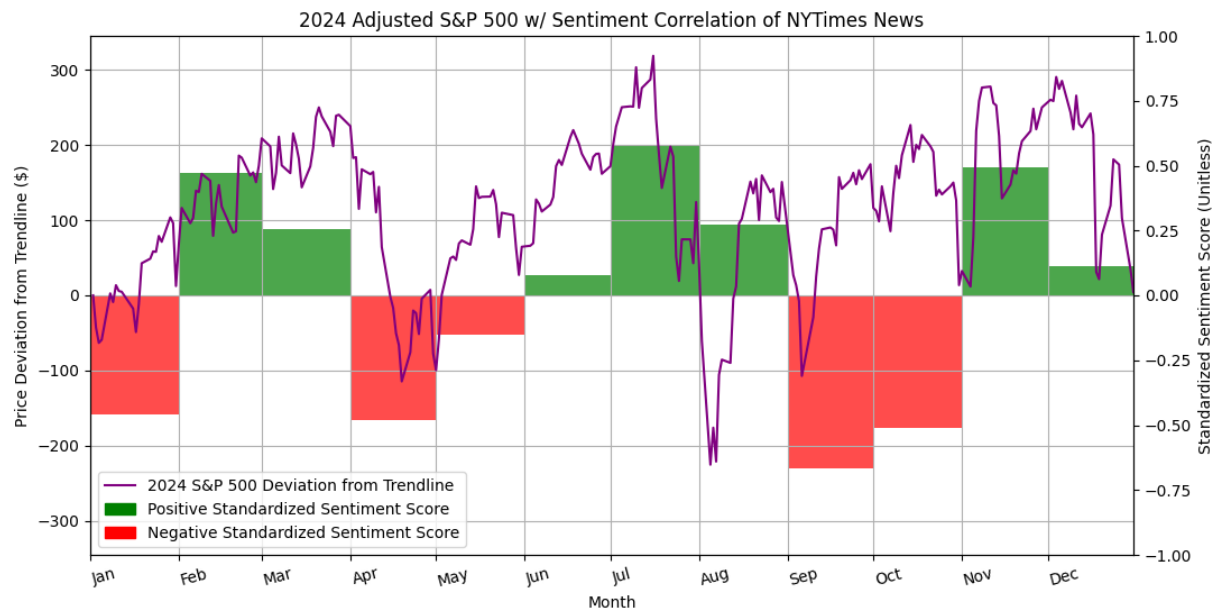




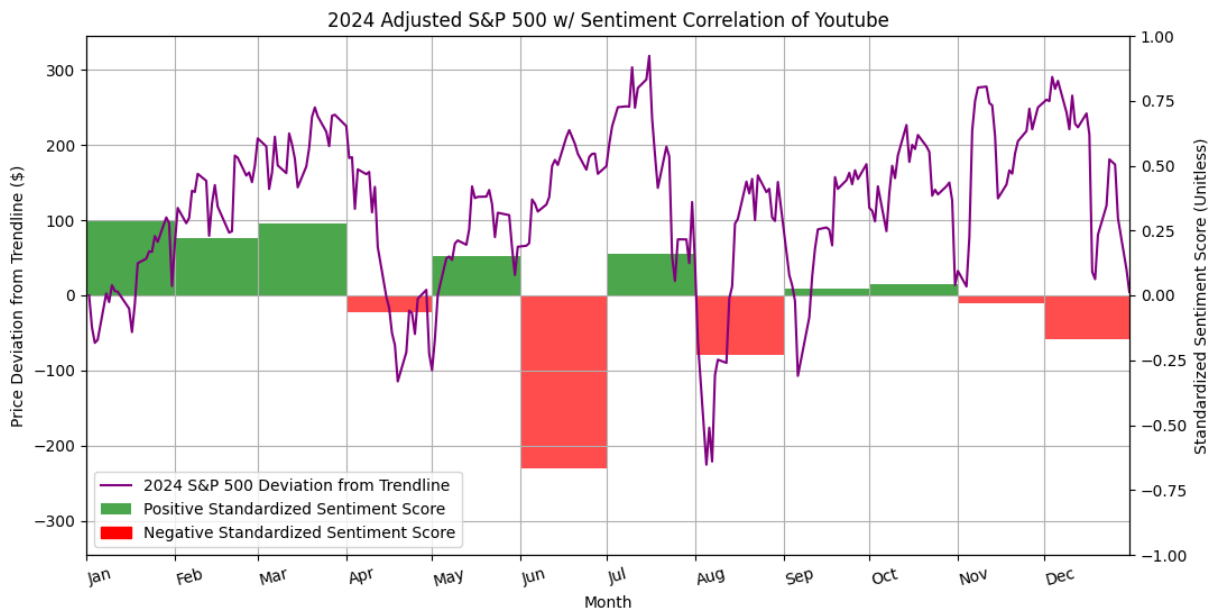
B Yahoo Finance Graph



C NYTimes Graph



D YouTube Comments Graph



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